STA 310: Homework 1

Instructions

- Write all narrative using full sentences. Write all interpretations and conclusions in the context of the data.
- Be sure all analysis code is displayed in the rendered pdf.
- If you are fitting a model, display the model output in a neatly formatted table. (The `tidy` and `kable` functions can help!)
- If you are creating a plot, use clear and informative labels and titles.
- Render and back up your work reguarly, such as using Github.
- When you're done, we should be able to render the final version of the Rmd document to fully reproduce your pdf.
- Upload your pdf to Gradescope. Upload your Rmd, pdf (and any data) to Canvas.

Exercises

Exercises 1 - 4 are adapted from exercises in Section 1.8 of @roback2021beyond.

Exercise 1

Consider the following scenario:

> Researchers record the number of cricket chirps per minute and temperature during that time. They use linear regression to investigate whether the number of chirps varies with temperature.

a. Identify the response and predictor variable.

   The predictor variable is the temperature, and the response variable is the number of chirps.

b. Write the complete specification of the statistical model.

   $y_i = \beta_1 x_i + \beta_0 + \epsilon_i$

   For the ith observation, y is the number of chirps, and x is the temperature. β_0 is the y-intercept, and β_1 is the slope, representing the average change in number of chirps for every 1 degree increase in temperature.

c. Write the assumptions for linear regression in the context of the problem.

   Linearity: There is a linear relationship between temperature and chirps.

   Independence: Each recording of cricket chirps must be independent of other data recordings.

   Normality: The residuals of the model should follow a normal distribution.

   Homoscedasticity: The number of chirps should have constant variance.

Exercise 2

Consider the following scenario:

> A randomized clinical trial investigated postnatal depression and the use of an estrogen patch. Patients were randomly assigned to either use the patch or not. Depression scores were recorded on 6 different visits.

a. Identify the response and predictor variables.

   Predictor variable: Estrogen patch usage (binary)

   Response variable: Depression score

b. Identify which model assumption(s) are violated. Briefly explain your choice.

   The assumption of independence was violated because each patient visited 6 different times, and each of their depression scores is likely dependent on the previous one. Multiple measurements from the same subject should not be treated as independent.

Exercise 3

Use the Kentucky Derby case study in Chapter 1 of *Beyond Multiple Linear Regression.*

a. Consider Equation (1.3) in Section 1.6.3. Show why we have to be sure to say "holding year constant", "after adjusting for year", or an equivalent statement, when interpreting β_2.

   The model is $Y_i = \beta_0 + \beta_1 Yearnew_i + \beta_2 Fast_i + \epsilon_i$, indicating that both Fast and Yearnew serve as predictors for winning speed, which is the response variable. Therefore, β_2 indicates the predicted change in winning speed when conditions are fast, but only assuming that Yearnew does not change..

b. Briefly explain why there is no error (random variation) term ϵ_i in Equation (1.4) in Section 1.6.6?

   This equation shows the estimated winning speeds, as indicated by $\hat{Y}_i$. The equation for expected real winning speeds does have error terms, but our estimations are fixed by our beta values.

Exercise 4

The data set `kingCountyHouses.csv` in the `data` folder contains data on over 20,000 houses sold in King County, Washington (@kingcounty).

We will use the following variables:

- `price` = selling price of the house
- `sqft` = interior square footage

See Section 1.8 of Beyond Multiple Linear Regression for the full list of variables.

a. Fit a linear regression model with `price` as the response variable and `sqft` as the predictor variable (Model 1). Interpret the slope coefficient in terms of the expected change in price when `sqft` increases by 100.

```r
houses <- read.csv("homeworks/data/kingCountyHouses.csv")
head(houses)
```

```
##      price date bedrooms bathrooms sqft floors waterfront yr_built yr_renovated
## 1  221900  285        3      1.00 1180      1          0     1955            0
## 2  538000  342        3      2.25 2570      2          0     1951         1991
## 3  180000  420        2      1.00  770      1          0     1933            0
## 4  604000  342        4      3.00 1960      1          0     1965            0
## 5  510000  413        3      2.00 1680      1          0     1987            0
## 6 1225000  131        4      4.50 5420      1          0     2001            0
```

```r
housemodel <- lm(price ~ sqft, data = houses)
summary(housemodel)
```

```
##
## Call:
## lm(formula = price ~ sqft, data = houses)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1476062  -147486   -24043   106182  4362067
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept) -43580.743   4402.690  -9.899   <2e-16 ***
## sqft           280.624      1.936 144.920   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 261500 on 21611 degrees of freedom
## Multiple R-squared:  0.4929, Adjusted R-squared:  0.4928
## F-statistic: 2.1e+04 on 1 and 21611 DF,  p-value: < 2.2e-16
```

The coefficient is approximately 280.62, representing the expected change in price in dollars for a 1 unit increase in square feet. Correspondingly, when square feet increases by 100, price is expected to increase by $28,062.

b. Fit Model 2, where `logprice` (the natural log of price) is now the response variable and `sqft` is still the predictor variable. How is the `logprice` expected to change when `sqft` increases by 100?

```r
housemodel2 <- lm(log(price) ~ sqft, data = houses)
summary(housemodel2)
```

```
##
## Call:
## lm(formula = log(price) ~ sqft, data = houses)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.97781 -0.28543  0.01472  0.26070  1.27628
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 1.222e+01  6.374e-03  1916.9   <2e-16 ***
## sqft        3.987e-04  2.803e-06   142.2   <2e-16 ***
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.3785 on 21611 degrees of freedom
## Multiple R-squared:  0.4835, Adjusted R-squared:  0.4835
## F-statistic: 2.023e+04 on 1 and 21611 DF,  p-value: < 2.2e-16
```

The new coefficient is 0.0003987, so, when square feet increases by 100, log(price) is expected to increase by \$0.03987.

c. Recall that $log(a) - log(b) = log(\frac{a}{b})$. Use this to derive how the `price` is expected to change when `sqft` increases by 100 based on Model 2.

$$
\begin{aligned}
\beta_0 + \beta_1(\text{sqft}) &= \log(\text{price}_1) \\
\beta_0 + \beta_1(\text{sqft} + 100) &= \log(\text{price}_2) \\
\log(\text{price}_2) - \log(\text{price}_1) &= \beta_1(\text{sqft} + 100) - \beta_1(\text{sqft}) \\
\log(\text{price}_2) - \log(\text{price}_1) &= 100\beta_1 \\
\log\left(\frac{\text{price}_2}{\text{price}_1}\right) &= 100\beta_1 \\
\frac{\text{price}_2}{\text{price}_1} &= e^{100\beta_1} = e^{100*0.0003987} \\
\frac{\text{price}_2}{\text{price}_1} &= e^{0.03987} = 1.0407
\end{aligned}
$$

When sqft increases by 100, price is expected to be multiplied by 1.0407.

d. Fit Model 3, where `price` and `logsqft` (the natural log of sqft) are the response and predictor variables, respectively. How does the price expected to change when sqft increases by 10%? *As a hint, this is the same as multiplying sqft by 1.10.*

```r
housemodel3 <- lm(price ~ log(sqft), data = houses)
summary(housemodel3)
```

```
##
## Call:
## lm(formula = price ~ log(sqft), data = houses)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -606252 -170067  -33139  106342 6183772
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) -3451377      35169  -98.14   <2e-16 ***
## log(sqft)     528648       4651  113.67   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 290400 on 21611 degrees of freedom
## Multiple R-squared:  0.3742, Adjusted R-squared:  0.3742
## F-statistic: 1.292e+04 on 1 and 21611 DF,  p-value: < 2.2e-16
```

$$\beta_0 + \beta_1 \log(\text{sqft}) = \text{price}_1$$
$$\beta_0 + \beta_1 \log(1.10 * \text{sqft}) = \text{price}_2$$
$$\text{price}_2 - \text{price}_1 = \beta_1 \log(1.10 * \text{sqft}) - \beta_1 \log(\text{sqft})$$
$$\text{price}_2 - \text{price}_1 = \beta_1 [\log(1.10 * \text{sqft}) - \log(\text{sqft})]$$
$$\text{price}_2 - \text{price}_1 = \beta_1 \log\left(\frac{1.10 * \text{sqft}}{\text{sqft}}\right) = \beta_1 \log(1.10)$$
$$\text{price}_2 - \text{price}_1 = 528648 * \log(1.10) = 528648 * 0.0953$$
$$\text{price}_2 - \text{price}_1 = 50380$$

When sqft increases by 10%, price is expected to increase by $50,380.

Exercise 5

The goal of this analysis is to use characteristics of 593 colleges and universities in the United States to understand variability in the early career pay, defined as the median salary for alumni with 0 - 5 years of experience. The data was obtained from TidyTuesday College tuition, diversity, and pay, and was originaly collected from the PayScale College Salary Report.

The data set is located in `college-data.csv` in the `data` folder. We will focus on the following variables:

variable	class	description
name	character	Name of school
state_name	character	state name
type	character	Public or private
early_career_pay	double	Median salary for alumni with 0 - 5 years experience (in US dollars)
stem_percent	double	Percent of degrees awarded in science, technology, engineering, or math subjects
out_of_state_total	double	Total cost for in-state residents in USD (sum of room & board + out of state tuition)

a. Visualize the distribution of the response variable `early_career_pay`. Write 1 - 2 observations from the plot.

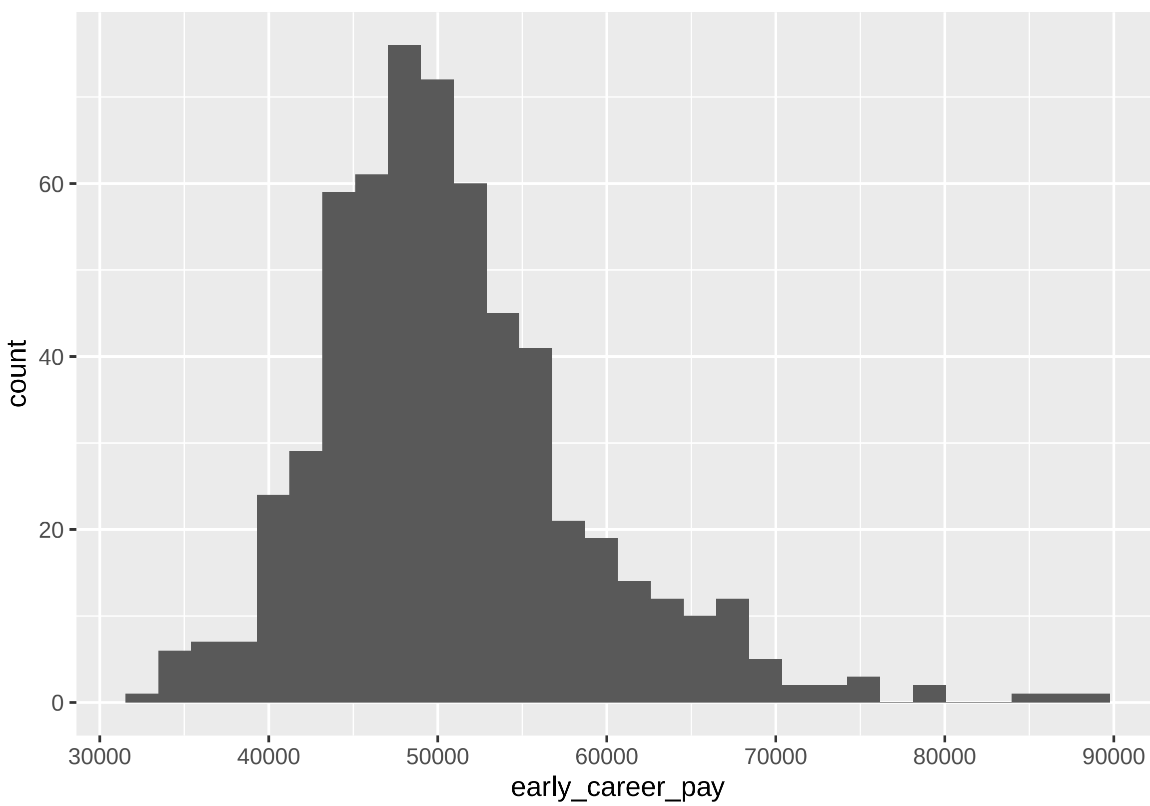
```r
colleges <- read.csv("homeworks/data/college-data.csv")
head(colleges)
```

```
##                                    name       state state_code    type
## 1                  Adams State University    Colorado         CO  Public
## 2 Adventist University of Health Sciences     Florida         FL Private
## 3                     Agnes Scott College     Georgia         GA Private
## 4                 Alabama State University     Alabama         AL  Public
## 5                Alaska Pacific University      Alaska         AK Private
## 6                 Albertus Magnus College Connecticut         CT Private
##   degree_length room_and_board in_state_tuition in_state_total
## 1        4 Year           8782             9440          18222
## 2        4 Year           4200            15150          19350
## 3        4 Year          12330            41160          53490
## 4        4 Year           5422            11068          16490
## 5        4 Year           7300            20830          28130
```

```
## 6        4 Year          13200             32060          45260
##   out_of_state_tuition out_of_state_total rank early_career_pay mid_career_pay
## 1                20456              29238   16            44400          81400
## 2                15150              19350   14            51600          89800
## 3                41160              53490   14            46000          83600
## 4                19396              24818   20            39800          71500
## 5                20830              28130    3            50300          90000
## 6                32060              45260   18            49700          85900
##   make_world_better_percent stem_percent
## 1                        56            3
## 2                        88            5
## 3                        57           26
## 4                        61           16
## 5                        67            6
## 6                        62            2
```

```r
library(ggplot2)
ggplot(data = colleges, aes(x = early_career_pay)) + geom_histogram()
```

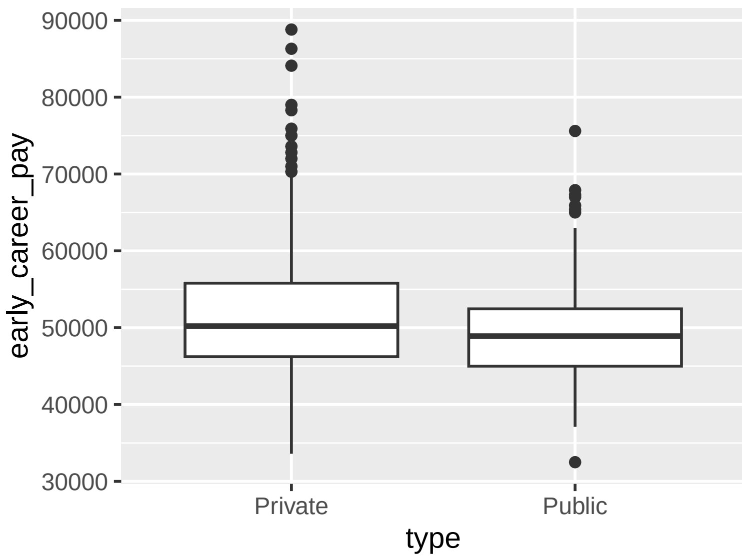
```
## `stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



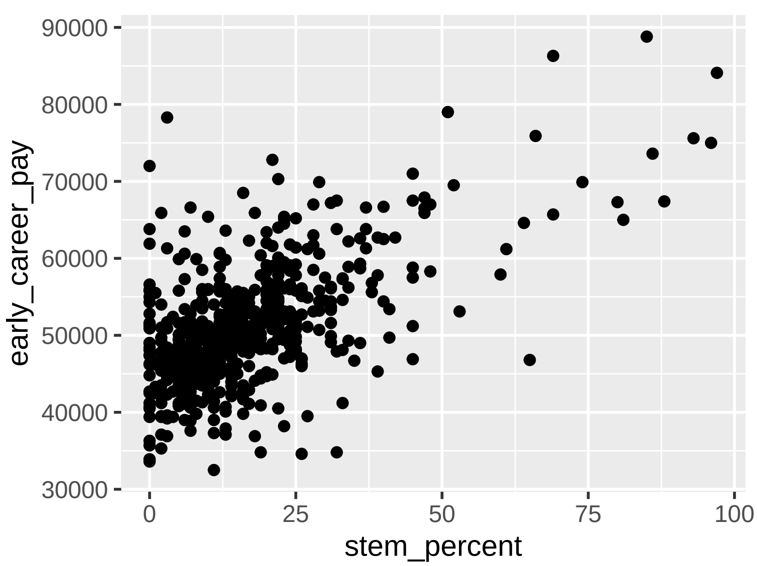
This histogram indicates that early career pay is right skewed, with a median payment around $50,000.

b. Visualize the relationship between (i) `early_career_pay` and `type` and (ii) `early_career_pay` and `stem_percent`. Write an observation from each plot.

```r
library(ggplot2)
ggplot(data = colleges, aes(y = early_career_pay, x = type)) + geom_boxplot()
```




```r
ggplot(data = colleges, aes(y = early_career_pay, x = stem_percent)) + geom_point()
```



The first graph indicates that private schools have a slightly higher median early career salary than public schools. Private schools' early career salaries also appear more right skewed than public schools, with a higher maximum.

The second graph demonstrates a positive relationship between the percent of degrees a university awards in STEM and its early career pay.

c. Below is the specification of the statistical model for this analysis. Fit the model and neatly display the results using 3 digits. Display the 95% confidence interval for the coefficients.

$$
\begin{aligned}
early\_career\_pay_i = \beta_0 + \beta_1\ out\_of\_state\_total_i + \beta_2\ type & \quad (1)\\
+ \beta_3\ stem\_percent_i + \beta_4\ type * stem\_percent_i & \quad (2)\\
+ \epsilon_i, \quad \text{where } \epsilon_i \sim N(0, \sigma^2) & \quad (3)
\end{aligned}
$$

```r
collegemodel <- lm(early_career_pay ~
                     out_of_state_total + type + stem_percent + type:stem_percent,
                   data = colleges)
summary(collegemodel)
```

```
##
## Call:
## lm(formula = early_career_pay ~ out_of_state_total + type + stem_percent +
##     type:stem_percent, data = colleges)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -14330  -3038   -413   2437  37028
##
## Coefficients:
##                          Estimate Std. Error t value Pr(>|t|)
## (Intercept)             3.622e+04  8.502e+02  42.598   <2e-16 ***
## out_of_state_total      2.530e-01  1.847e-02  13.692   <2e-16 ***
## typePublic              1.185e+03  7.688e+02   1.541    0.124
## stem_percent            2.143e+02  1.930e+01  11.104   <2e-16 ***
## typePublic:stem_percent 4.954e+01  3.387e+01   1.462    0.144
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 5440 on 588 degrees of freedom
## Multiple R-squared:  0.5324, Adjusted R-squared:  0.5292
## F-statistic: 167.4 on 4 and 588 DF,  p-value: < 2.2e-16
```

```r
confint(collegemodel, level = 0.95)
```

```
##                                2.5 %       97.5 %
## (Intercept)             34547.8624888 3.788755e+04
## out_of_state_total          0.2166625 2.892315e-01
## typePublic               -324.8132830 2.694853e+03
## stem_percent              176.4015255 2.522112e+02
## typePublic:stem_percent   -16.9920337 1.160687e+02
```

d. How many degrees of freedom are there in the estimate of the regression standard error σ?

```r
degrees <- nrow(colleges) - 5 # four parameters in model plus intercept
degrees
```

```
## [1] 588
```

```r
df.residual(collegemodel) # confirming calculation
```

```
## [1] 588
```

There are 588 degrees of freedom, calculated by n - p - 1.

e. What is the 95% confidence interval for the amount in which the intercept for public institutions differs from private institutions?

   The confidence level output for typePublic is (-324.813, 2694.853), indicating how much the intercept for public institutions differs from private. We are 95% confident that the intercept for public institutions is between $325 lower and $2695 higher than the intercept for private, all else constant.

Exercise 6

Use the analysis from the previous exercise to write a paragraph (~ 4 - 5 sentences) describing the differences in early career pay based on the institution characteristics. *The summary should be consistent with the results from the previous exercise, comprehensive, answers the primary analysis question, and tells a cohesive story (e.g., a list of interpretations will not receive full credit).*

This analysis is examining 593 American colleges and universities to understand variability in their early career pay, or the median salary for their alumni with less than 5 years of experience. Initial exploratory data analysis indicated that private schools had a higher median early career salary than public schools, with a more right-skewed distribution and higher maximum values. We fit a regression model accounting for institutional type, STEM focus, and tuition to analyze these factors. After controlling for these other characteristics, we are 95% confident that public school graduates earn between $325 less and $2,695 more than their private school counterparts. These results suggest that institutional type alone explains little of the variation in early career outcomes.

Grading

Total	**50**
Ex 1	8
Ex 2	4
Ex 3	7
Ex 4	12
Ex 5	12
Ex 6	4
Workflow & formatting	3

The “Workflow & formatting” grade is to based on the organization of the assignment write up along with the reproducible workflow. This includes having an organized write up with neat and readable headers, code, and narrative, including properly rendered mathematical notation. It also includes having a reproducible Rmd/Quarto document that can be rendered to reproduce the submitted PDF.